

ANNISA SELVIRA RA'INA GUSTIYANINGSIH

Electrical Engineer | Entrepreneur | Operations & Control Systems Specialist

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PROFESSIONAL SUMMARY

Electrical Engineer (Universitas Telkom, GPA 3.53/4.00) and entrepreneur combining hands-on expertise in control systems, embedded hardware, and renewable energy simulation with proven business leadership. Founded and scaled a freight logistics company to 7 vehicles and 1,000+ clients with full P&L ownership. Led a 6-person team to 1st Place at KRI 2023 Regional 1 and authored a peer-reviewed publication on hybrid power generation for remote island electrification.

TECHNICAL SKILLS

Control & Automation: PLC Omron, MATLAB/Simulink, DigSILENT PowerFactory, HOMER, SCADA, Proteus/PSPICE

Embedded & Hardware: Arduino, STM32, ESP8266, PCB Design (Autodesk Eagle, SMD/Non-SMD)

Programming & Data: C/C++, Python (Pandas, NumPy, Scikit-learn, Matplotlib), MySQL, Git

Renewable Energy: QBlade, PVsyst, HOMER, On-Grid & Off-Grid Solar PV Design

Design & Modeling: SolidWorks, Autodesk Inventor, Figma, Blender 3D

Business & Leadership: P&L Management, SOP Development, Fleet Operations, KPI Monitoring, Team Leadership

Languages: Bahasa Indonesia (Native), English (Professional Working Proficiency)

PROFESSIONAL EXPERIENCE

Founder & Operations Manager | Jasa Transport Selwi, Kalimantan Tengah

Mar 2023 – Present

- Founded and scaled a freight logistics business from zero to 7 vehicles, 7 drivers, and 1,000+ clients across Central Kalimantan within 3 years, with full P&L ownership.
- Manage daily operations route scheduling, fleet maintenance, and driver deployment sustaining on-time delivery with zero critical service failures.
- Designed operational SOPs for driver coordination and client communication, improving service consistency through disruptions such as breakdowns and route changes.

Branch Coordinator & Teacher | Coding Bee Academy, Kab. Bandung Barat

May 2024 – Mar 2026

- Managed branch operations for 5+ educators and 50+ students; secured partnerships with 3+ schools, driving a 20% enrollment increase within 12 months.
- Implemented enrollment-tracking and coaching systems, sustaining 85%+ student retention and reducing staff turnover.
- Designed and delivered project-based curricula in robotics and back-end web development, raising project completion rates by ~25%.

Research Laboratory Assistant | EIRRG, Universitas Telkom, Bandung

Oct 2021 – Oct 2023

- Executed PCB schematic capture and SMD layout across 5+ robotics prototypes with zero critical layout errors.
- Streamlined component procurement tracking, cutting lead times by ~30% and improving R&D budget transparency.
- Supported firmware debugging and sensor integration testing for published autonomous-systems research.

Engineering Intern | PT Telkom Indonesia (Persero) Tbk, Kotawaringin Timur

Jul 2022 – Sep 2022

- Analyzed 10+ ATS panel diagrams and live voltage-flow data, informing maintenance strategy and contingency planning.
- Documented relay protection configurations for 5+ panel systems, supporting knowledge transfer and operational continuity.

KEY PROJECTS

Multi-Sensor Industrial Fire Safety System — Personal Project

Dec 2025 – Feb 2026

- Built an Arduino-based fire safety system with dual flame/temperature sensors, tiered alarm logic, and a secure reset interlock; designed and validated on physical hardware.

Hybrid Power Generation Feasibility Study — Karimunjawa Island (Capstone)

Aug 2022 – Aug 2023

- Led techno-economic analysis of 4 hybrid configurations; optimal design cut generation cost from Rp1,326 to Rp1,231/kWh, resulting in a peer-reviewed publication on remote island electrification.

Machine Learning Handwritten Letter Recognition (EMNIST)

Sep 2022 – Feb 2023

- Built an end-to-end classification pipeline on 145,600 samples using Scikit-learn, achieving reliable recognition across all 26 letter classes.

Solar PV On-Grid System Design & Economic Analysis

Sep 2022 – Feb 2023

- Designed a residential PLTS system for Jakarta (6.40 MWh/year, 79.42% performance ratio); confirmed feasibility via LCC, LCOE, NPV, and payback analysis.

EDUCATION

Bachelor of Engineering, Electrical Engineering | Universitas Telkom, Bandung

Aug 2019 – Aug 2023

- GPA: 3.53/4.00 | Specialization: Control Systems, Power Systems, Automation & Smart Energy Technologies.
- 1st Place Regional 1, Indonesian Robot Contest (KRI 2023), KRSRI Division; capstone research published in a peer-reviewed journal.

ACHIEVEMENTS & PUBLICATIONS

- Peer-reviewed publication: "Wind Turbine Design for a Power Generation System with Competitive Generation Costs on Karimunjawa Island."
- 1st Place, Indonesian Robot Contest (KRI 2023), KRSRI Division led a 4-person team building an autonomous rescue robot.

CERTIFICATIONS & TRAINING

- Fundamental AI & Coding Training (3-month program, May 2025) — Kementerian Pendidikan Dasar dan Menengah.
- Python for Data Professional: Beginner → Visualization → EDA — DQLab Bootcamp (Aug 2021).
- Exploratory Data Analysis, Data Visualization, Fundamental SQL, R Fundamental — DQLab Bootcamp (Aug 2021).